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Traffic Congestion Tracing and Evolution Analysis Based on Simulated Individual Trajectory Data

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Abstract

Background
With rapid urbanization, traffic congestion has become a major challenge in cities worldwide. The growing availability of trajectory data enables fine-grained analysis, but challenges remain in data completeness, modeling accuracy, and intelligent analysis.

Project Objective
This project aims to construct spatio-temporal models to trace and predict the congestion in urban road networks. Specifically, we do the following steps:

- Identify high-impact vehicles and trajectory patterns that contribute to congestion
- Simulate microscopic vehicle routing behavior and its impact on congestion
- Predict short-term congestion evolution at the road segment level

Project Highlight

Congestion Contribution Quantification Model
Build a congestion contribution quantification model based on individual trajectories to accurately identify high-pressure OD areas.
→ Results can be applied support the selection of new road locations.

Temporal and Spatial Variation Characteristics
Analyze the temporal and spatial variation characteristics of road usage and establish a road classification and prediction system.
→ Results can provide important reference for the intelligent planning of urban.

Mathematical Modeling with Actual Travel Patterns
Integrate traffic flow theory with large-scale trajectory data to conduct multi-dimensional and multi-level congestion tracing analysis.
→ Combine mathematical modeling with actual travel patterns to enhance the scientific and practical effectiveness of city traffic management.

Data & Exploratory Data Analysis

Data Preprocessing: Construct 10-minute interval road traffic dataset by combining road information with 24-hour vehicle trajectory data.
Fundamental analysis: According to the road classification, conduct basic analysis of information such as length and traffic capacity.
Basic Pattern: The basic commuting direction is northeast-southwest. During the morning and evening rush hours (7:00-10:00) and (18:00-22:00), there is a sudden increase in traffic volume.

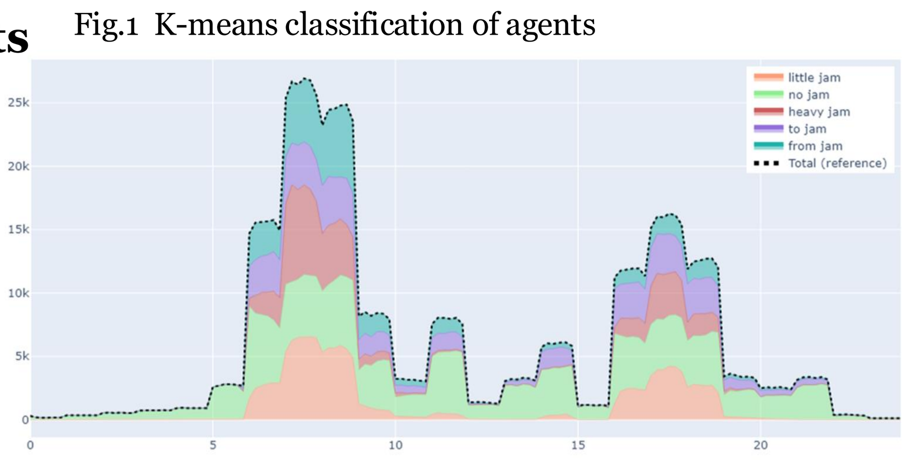
Methodology & Result

Congestion pattern classification

1. Classify the congestion status of agents

- Determine three parameter indicators.
- Use **k-means** to classify the congestion status of agents.(Fig.1)

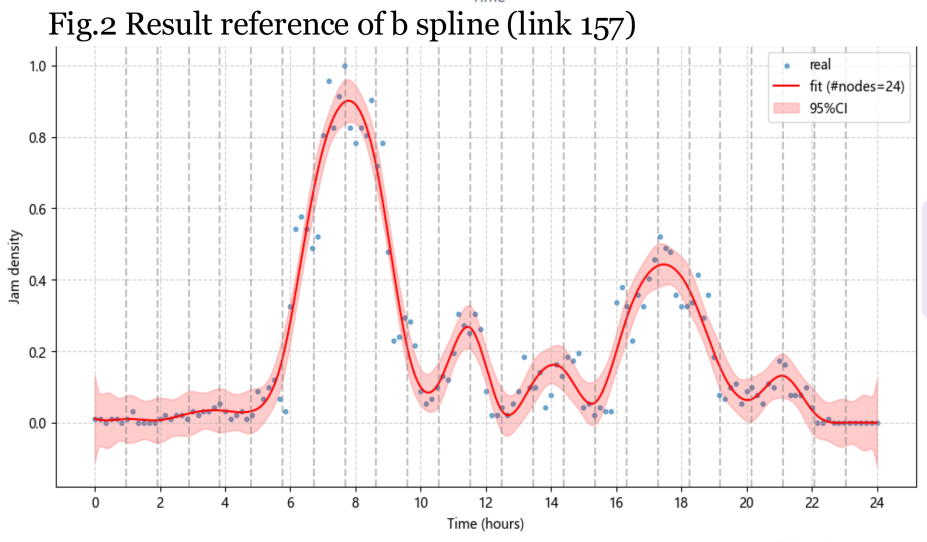
Fig.1 K-means classification of agents



2. Classify road congestion conditions

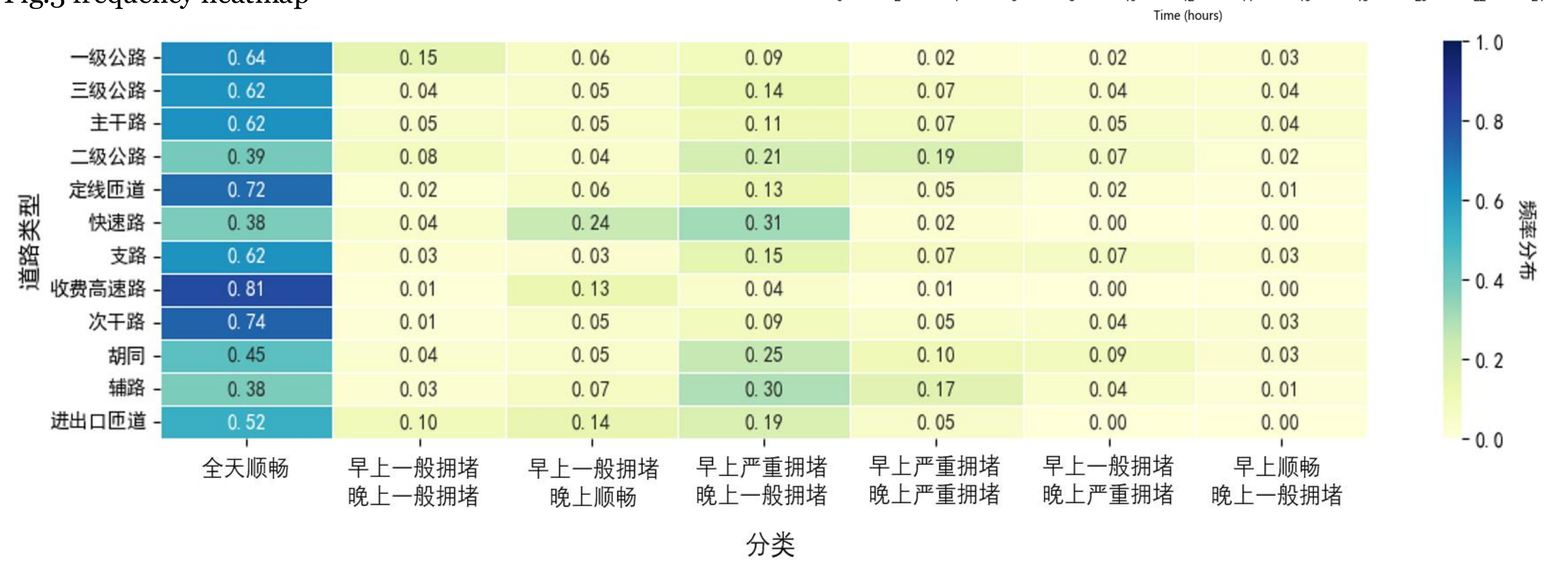
- For each road, divide by its maximum traffic congestion density and perform **B-spline** over time. (Fig.2)
- Use k-means to classify the trends in traffic congestion density changes.
- Combine the traffic congestion density trend categories with the maximum traffic congestion density for detailed classification.(Fig.4)

Fig.2 Result reference of b spline (link 157)



3. Analyze road congestion categories and road characteristics(Fig.3)

Fig.3 frequency heatmap



- Use a **random forest model** to predict road types based on road congestion categories, road length, and road start/end points.

Fig.4 Classification result example - ‘早上顺畅，晚上一般拥堵’

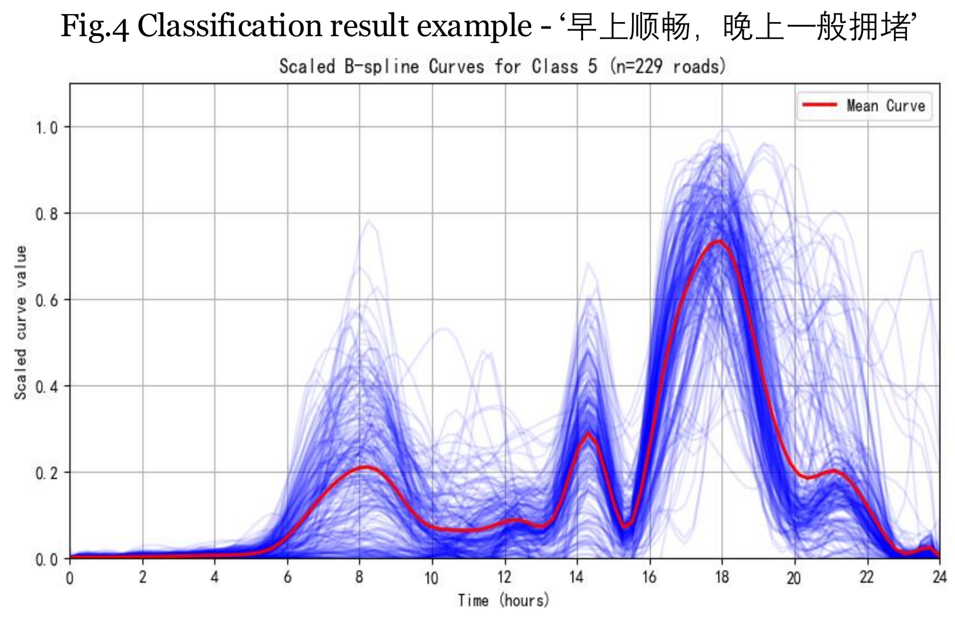
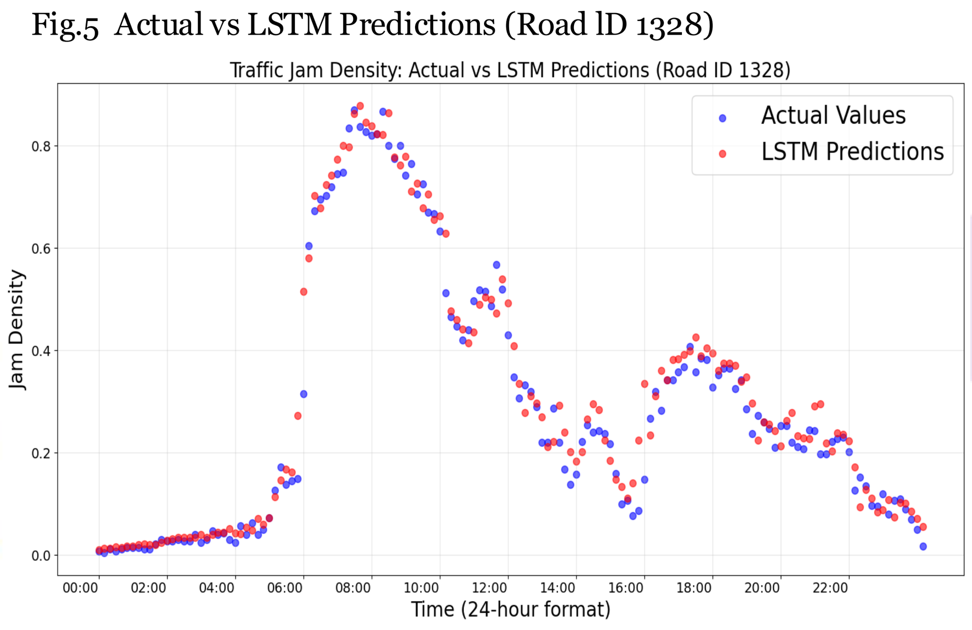


Fig.5 Actual vs LSTM Predictions (Road lD 1328)



Congestion spatial accounting

1. Short-term prediction of road congestion density
To predict the congestion density y_t at time t, use both static features of each link and recent congestion history.

- **Static Features:** A 13-dimensional vector describing fixed attributes of the road (e.g., length, number of lanes).
- **Dynamic Features:** Congestion densities from the past 30, 20, and 10minutes.

Combine one historical congestion value with the static features, forming a 14-dimensional input. Three inputs together form a sequence of shape 3×14, which is fed into an **LSTM** to predict the current congestion y_t.(Fig.5)

2. Queuing dynamics of traffic network nodes

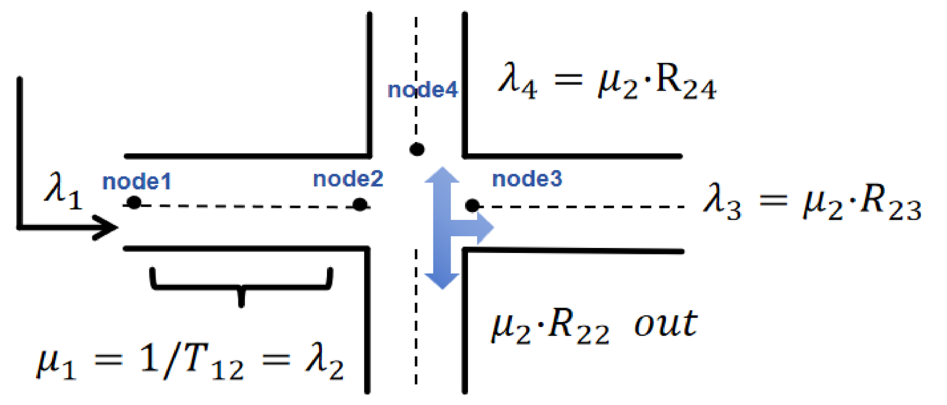
- Apply **queuing model** to the road network, construct **Continuous Markov Chain** to evolve the stochastic process; Calculate steady-state separately.
- Calculate average sojourn time a vehicle spent in the system **W**, average queuing time **Wq**, average queue length **Q**.

$$W = \frac{1}{\lambda}\cdot\sum_{x\in S}\pi_x . x \qquad W = W_q + \frac{1}{\mu}$$

$$Q = \sum_{x\in S}\pi_x . x - \frac{\lambda}{\mu}$$

λ: rate of poisson distribution to model arrival process
μ: rate of exponential distribution to model the time between occurrences of departures

Fig.6 Schematic diagram of queuing model



3. Quantitative Modeling of Congestion Contribution
Given the vehicle flow rate **F**, number of lanes **N**, and per-line capacity **C**, the congestion level of the roadway is defined as:

$$\alpha = \frac{F}{N\times C}$$

For a trip passing through n segments of length **Lj** with congestion level **αj** . The congestion contribution of the trip **Γ** is defined as:

$$\Gamma = \Sigma_{j=1}^{n} L_j \alpha_j$$

The research area is divided into **M × N** grids. For grid *(i, j)*, let **K** be the number of trips starting from or ending at that unit. The average congestion distributions are:

$$JamFrom_{\ i,j} = \frac{1}{K}\Sigma_{k=1}^{K}\Gamma_k^{from} \qquad JamTo_{\ i,j} = \frac{1}{K}\Sigma_{k=1}^{K}\Gamma_k^{to}$$

Fig.7 Congestion contribution of origin points

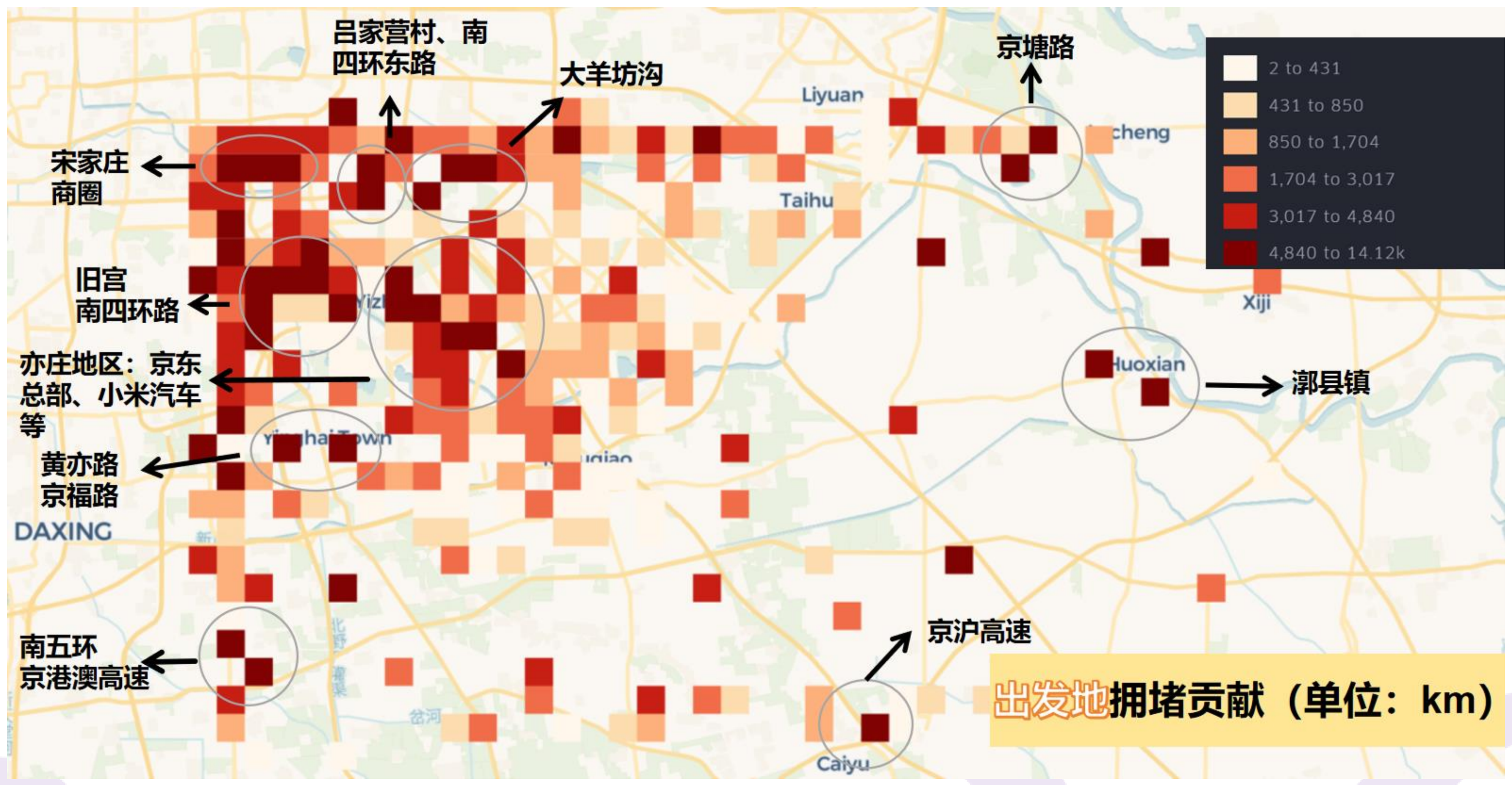
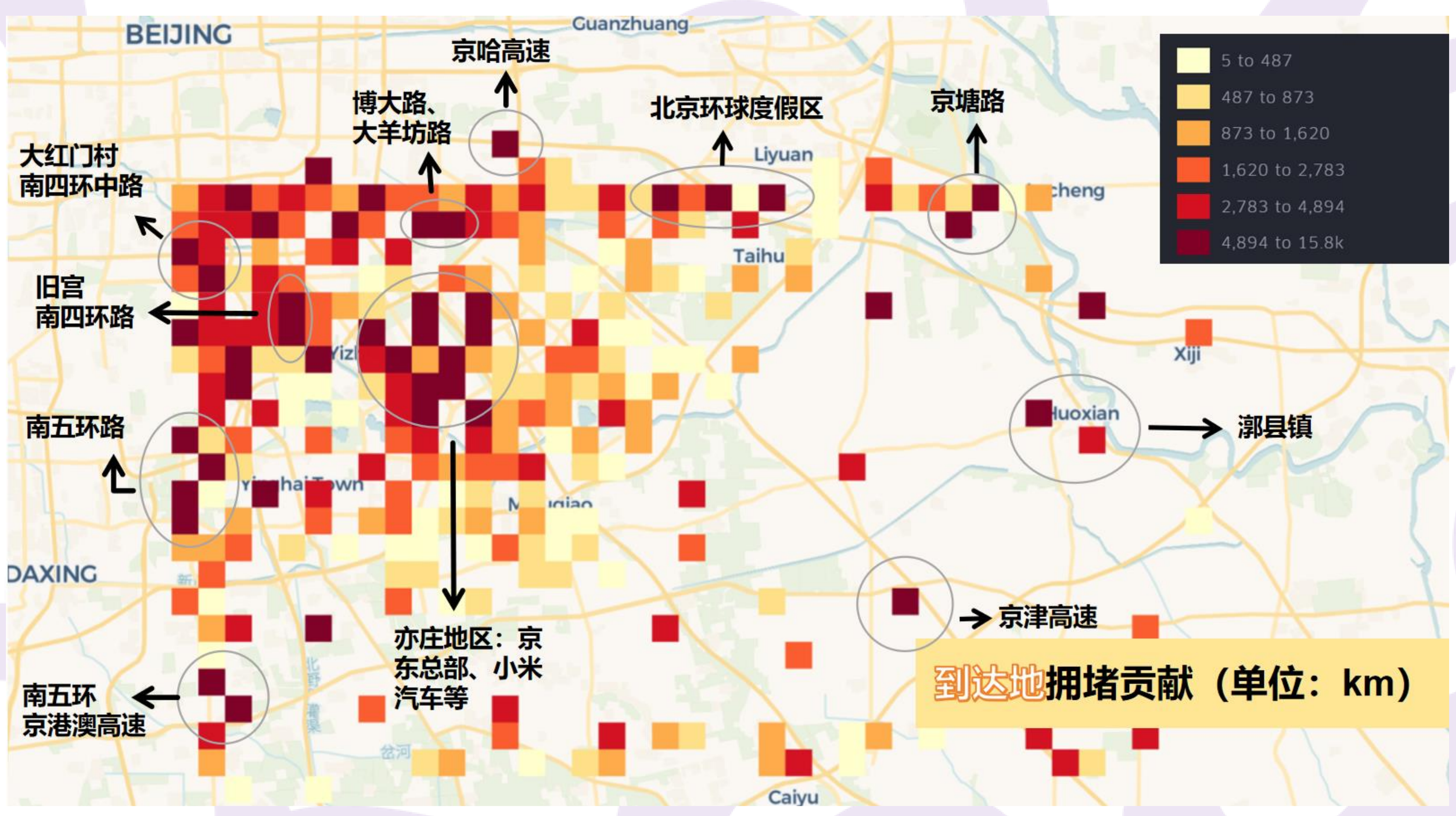


Fig.8 Congestion contribution of destination points



Summary